

DEVELOPMENT OF FACE

Dr. Ume Hani

Department of Oral Biology

4-week embryo:

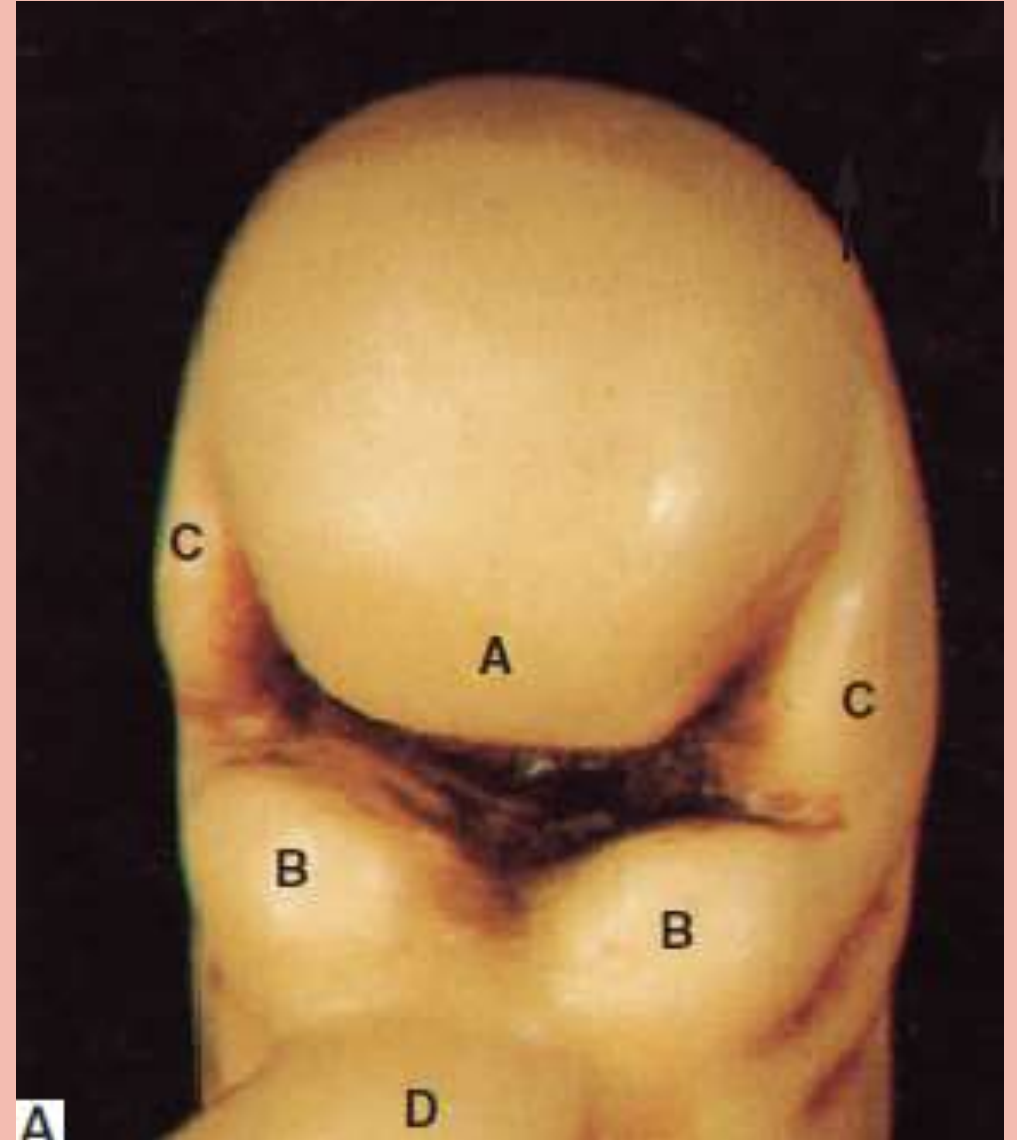
The five mesenchymal prominences can be recognised.

A: The frontonasal prominence, a slightly rounded elevation cranial to the stomodeum.

C: The maxillary prominences (dorsal portion of the 1st arch), lateral to the stomodeum.

B: The mandibular prominences, caudal to the stomodeum.

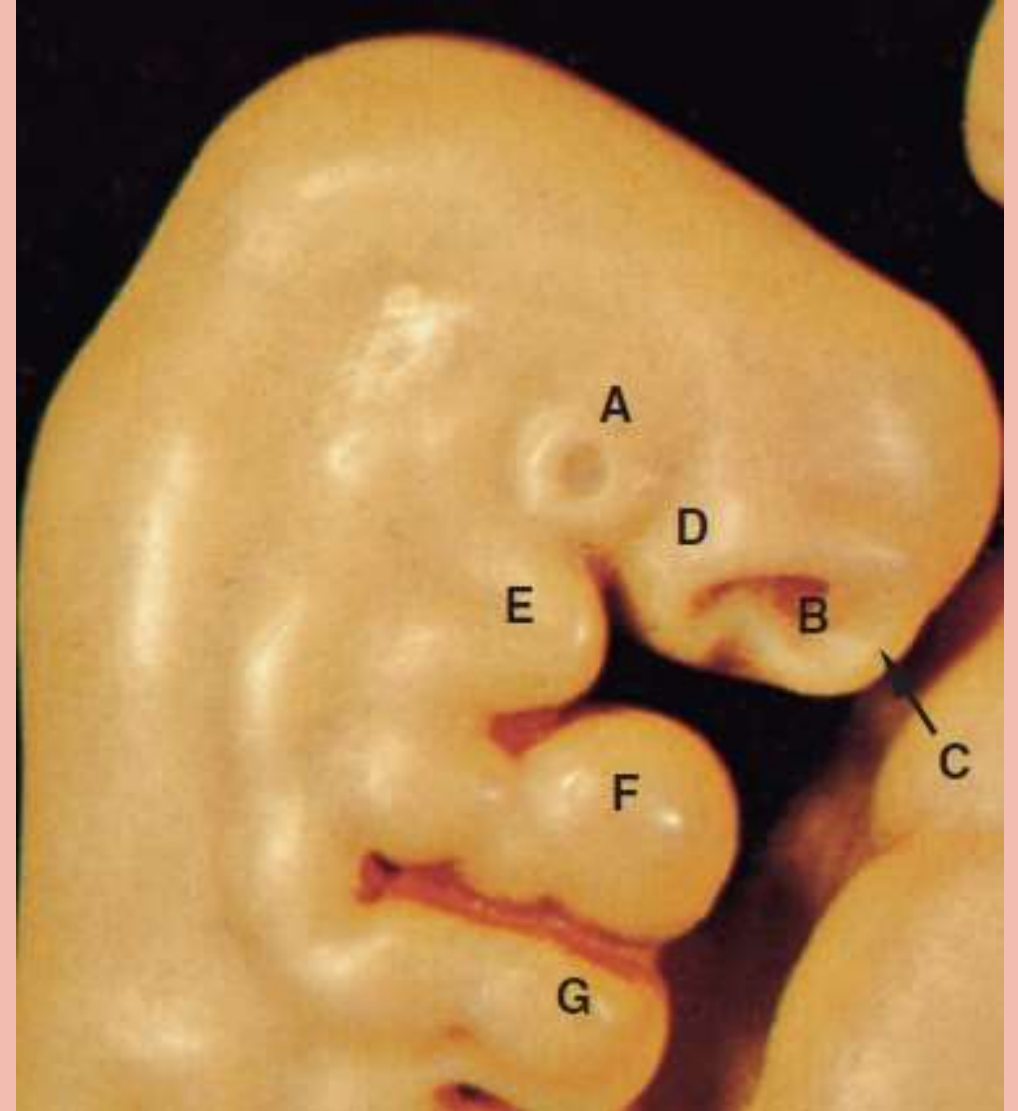
Development of the face is later completed by the formation of the nasal prominences.



5th-week embryo:

- The ectodermal thickenings give rise to the nasal (olfactory epi) and optic placodes (lens of the eyes).
- The nasal placodes sink into the underlying mesenchyme, creating two blind-ended nasal pits.
- Proliferation of mesenchyme from the frontonasal process around the openings of the nasal pits produces the medial and lateral nasal processes.

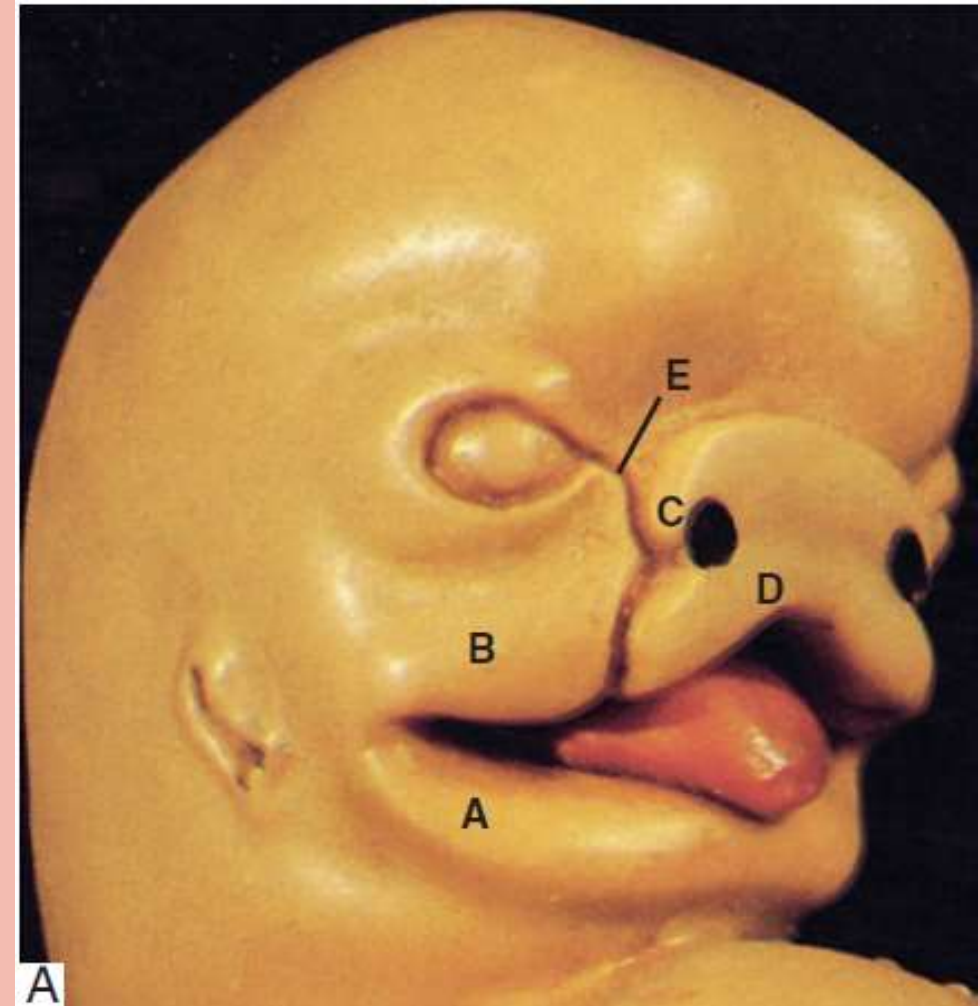
A: optic placode, B: nasal pit, C: = medial nasal process; D: lateral nasal process; E: maxillary process; F: mandibular process.



6th-week embryo:

- The nasal pits deepen and form nasal sacs. The two mandibular processes fuse in the midline to form the tissues of the lower jaw.
- The mandibular processes and maxillary processes form angles of the mouth.
- Medial nasal processes will form intermaxillary segment.
- Lateral nasal processes will form alae of nose.

A = mandibular process; B = maxillary process; C = lateral nasal process; D = medial nasal process; E = naso-optic furrow.



Nasolacrimal ducts:

Between the merging maxillary and the lateral nasal processes lie the naso-optic furrows.



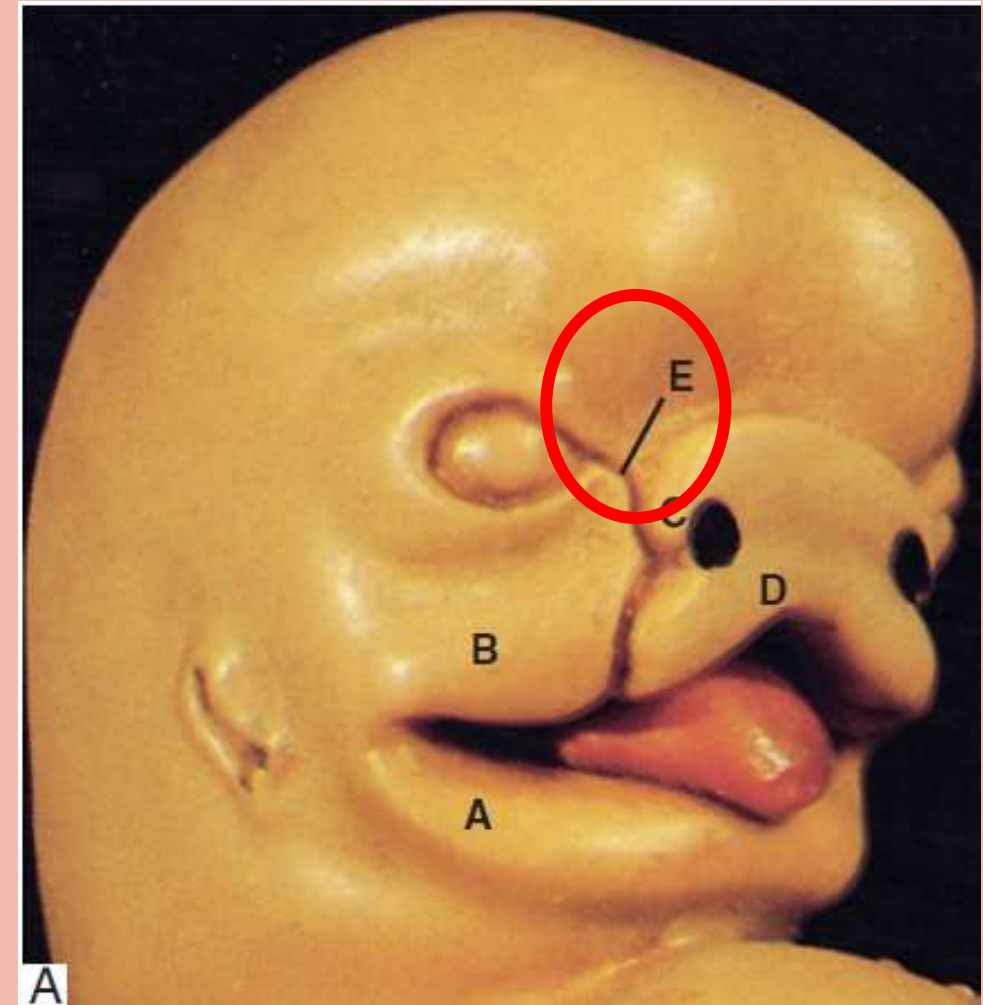
Ectodermal rod like thickening in the nasolacrimal groove.



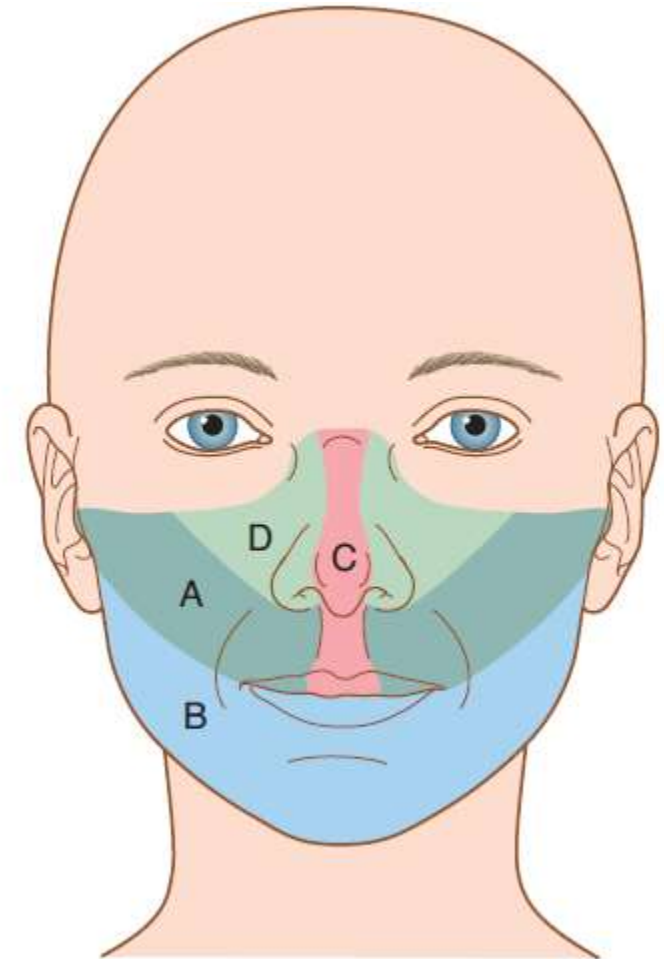
This solid cord of cells separates from the surface ectoderm and lies in the underlying mesenchyme.



The cord gets canalized to form the nasolacrimal duct.

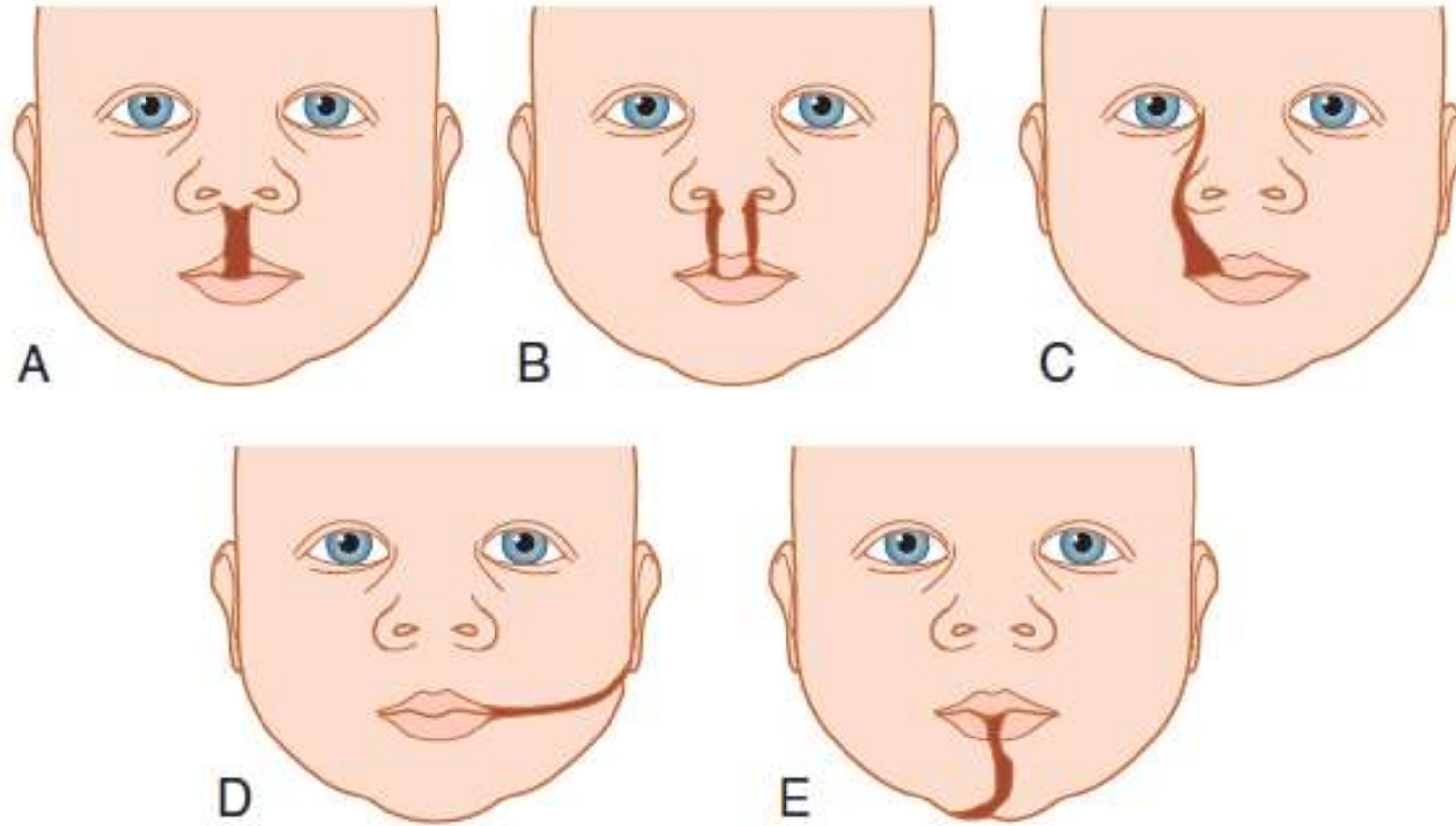


Frontonasal process	→	Forehead, bridge of the nose, medial and lateral nasal processes
A- Maxillary process	→	Cheeks, lateral portion of upper lip
B- Mandibular process	→	Lower jaw, lower lip
C- Medial nasal process	→	Intermaxillary segment: philtrum of upper lip, tip of nose, primary palate, premaxilla with 4 incisors
D- Lateral nasal process	→	Alae of nose



**SYNDROMES
AND
CLINICAL
CONSIDERATIONS**

FACIAL CLEFTS



HOLOPROSENCEPHALY:

This is a congenital abnormality in which the developing forebrain fails to divide into two separate hemispheres. It can be a numerical chromosomal abnormality (syndromic hydroprosencephaly) or a gene mutation (nonsyndromic hydroprosencephaly).

Facial defects:

1. Cyclopia
2. absence or severe reduction of nasal structures
- microcephaly
3. cleft palate and cleft lip.

Maternal/environmental risk factors:

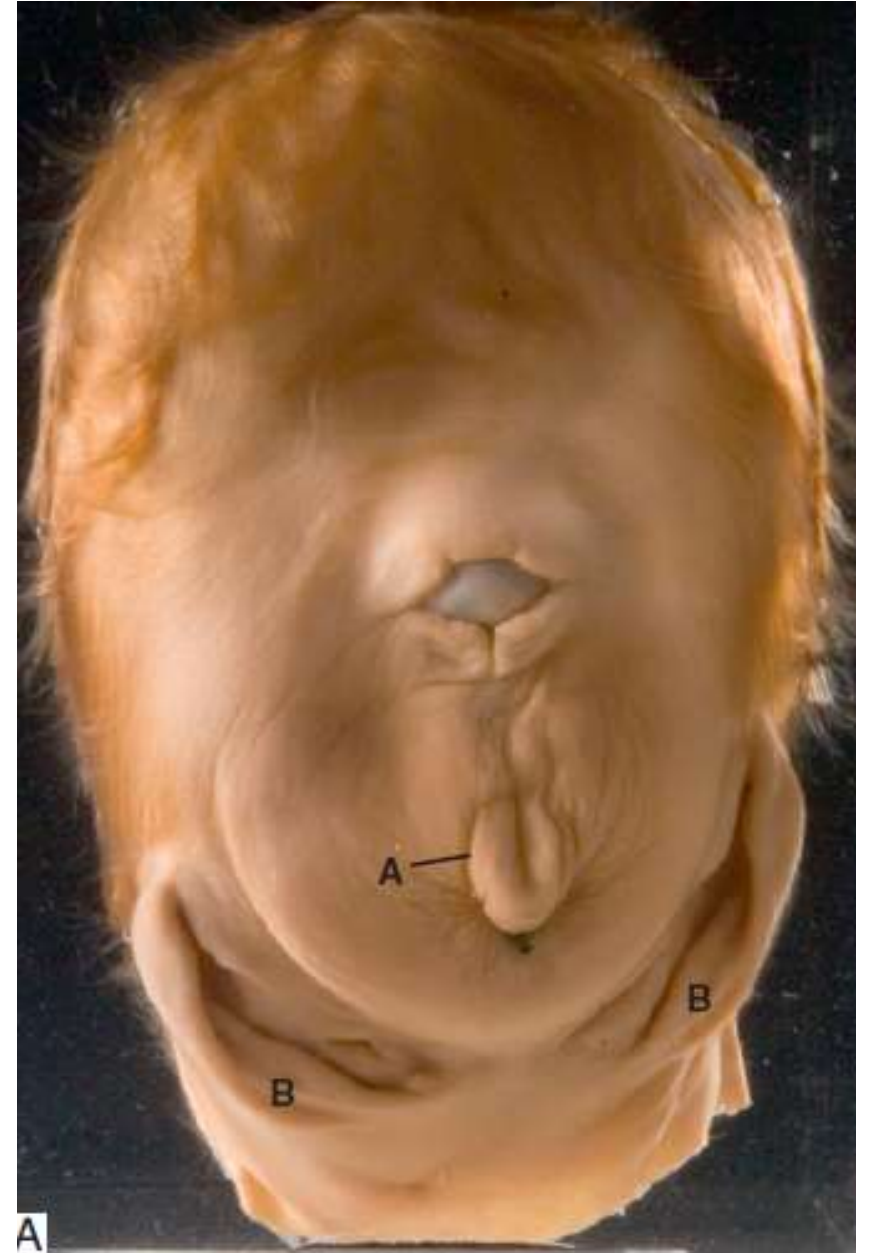
Maternal diabetes mellitus

Maternal infections (syphilis, rubella)

Retinoic acid exposure

Alcohol

Anticonvulsants

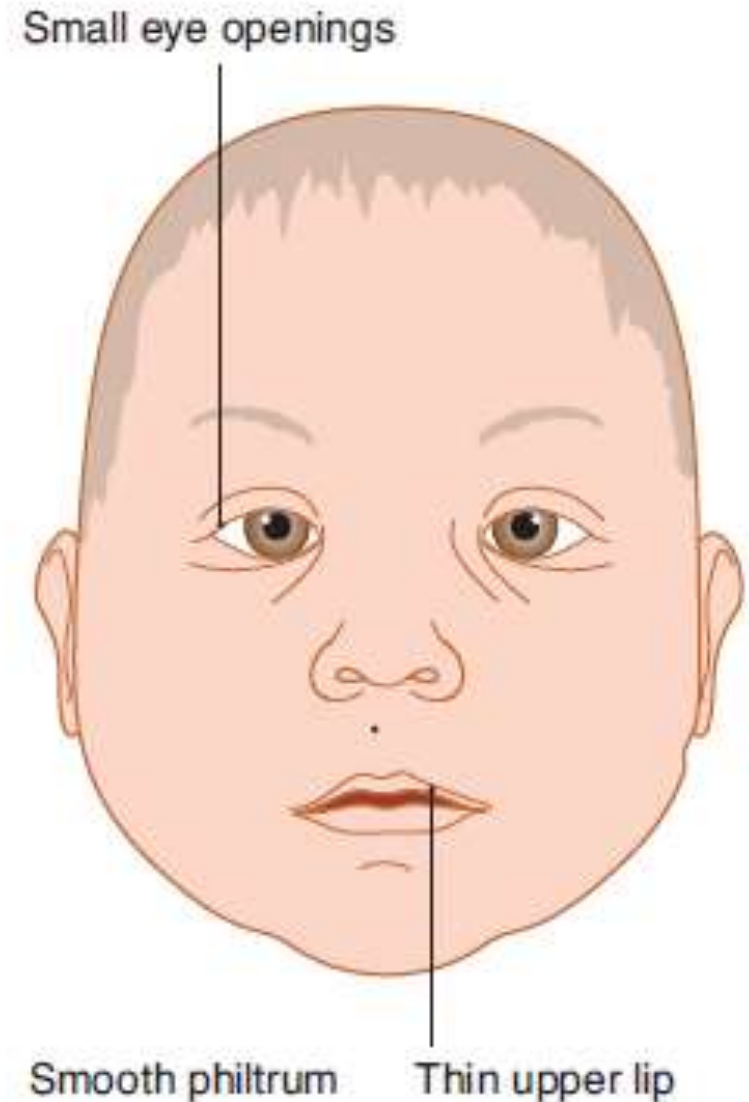


FETAL ALCOHOL SPECTRUM DISORDER

Caused by the mother drinking more than four standard units of alcohol per day during pregnancy.

Clinical Features:

1. Thin vermilion of the lips
2. Smooth, flattened philtrum
3. Small palpebral fissures: decreased eye width
4. Microcephaly
5. Low bridge of nose
6. Micrognathia
7. Mental instability



Both the Pierre Robin's Sequence and Treacher Collin's Syndrome are first arch syndromes. They affect all or most of the structures derived from the first pharyngeal arch (i.e., the mandibular and maxillary processes). First arch syndromes/sequences are caused either by insufficient migration of neural crest cells into the region of the first pharyngeal arch during week 4 of intrauterine life, by decreased cell proliferation or by increased cell death.

PIERRE ROBIN'S SEQUENCE:

A heterogeneous or autosomal recessive hereditary condition.

Clinical Features: (triad)

1. Micrognathia (underdeveloped mandible),
2. Glossoptosis (abnormal downward or backward placement of the tongue)
3. Cleft lip and palate

Difficulties:

1. Feeding difficulties
2. Airway obstruction and breathing problems
3. Speech problems due to cleft palate



TREACHER COLLIN'S SYNDROME

The disorder is an autosomal dominant condition. It happens because of the failure of migration of neural crest cells to facial region.

Clinical Features:

1. Downward-slanting eyes
2. Coloboma of eye
3. Micrognathia
4. Absent or malformed ears
5. Cleft lip and palate
6. Hypoplasia of the zygomatic bones

